Simulation of Multiphase Flow and Transport in Partially Melted Materials

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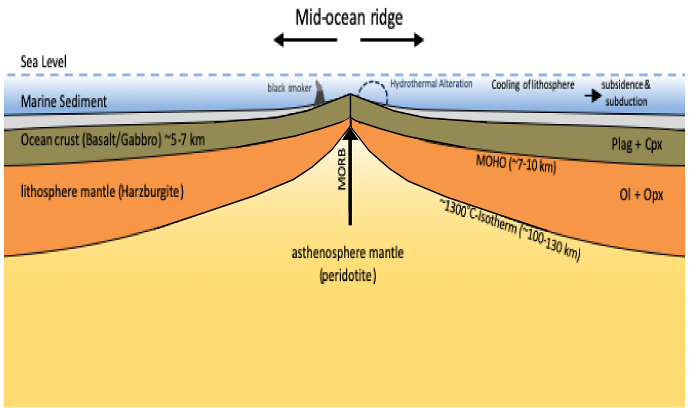
Partial Melting

1. Generates the oceanic and continental crust.
2. Determines the large-scale chemical evolution of the Earth.
3. Similar process may occur on other terrestrial planets.



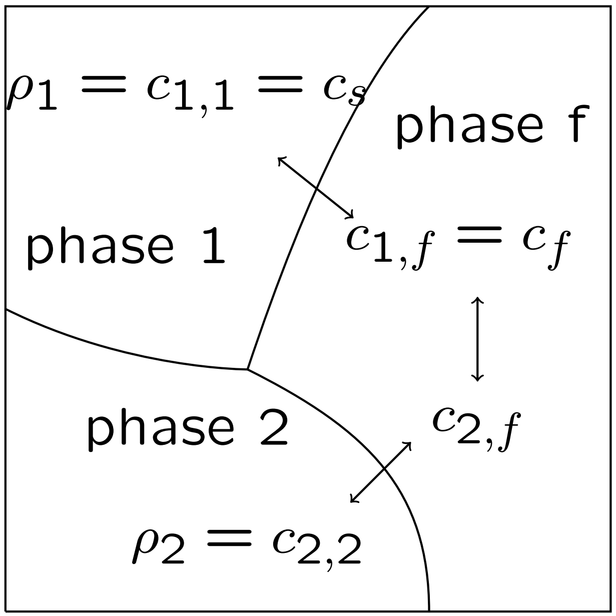
(Mid-ocean ridge, Wikipedia)

Goals.

- Refine mathematical model governing the system.
- Develop a new code for simulation of partial melting process.
    - New mixed finite element space.
    - Implement implicit WENO-AO method.
    - Eutectic binary phase behavior.
- Perform numerical simulation help understanding of physics.

1. The Equations Governing Partially Melting System

Elementary Representative Volume.



Mixture theory for Mantle Dynamics

(McKenzie 1984) Both fluid (f) and solid (s) phases exist at each point of the domain Ω.
Define the mixture variable as:

$$\overline{\Psi} = \phi\Psi_f + (1-\phi)\Psi_s = \Psi_s + \phi\Psi_r,$$

$$\Psi_r = \Psi_f - \Psi_s.$$

Governing equations

1. Darcy for fluid melt. Fluid Melts between rock crystals, forming a porous medium.
2. Stokes for solid rock. Highly viscous creep of the solid mantle.
3. Transport for thermodynamics. Transport energy and mass.
4. Phase behavior. Approximated with binary eutectic phase behavior.

Concentration of Components

Let $c_{i,j}$ denote the concentration of component i in phase j. $M_{i,j}$ and $V_{i,j}$ are the mass and volume of component $i \in \{1, 2\}$ in phase $j \in \{1, 2, f\}$.

Within Phase 1.

$$c_s = c_{1,1} = \rho_1 = \frac{M_{1,1}}{V_1}, \qquad c_{2,1} = 0 = M_{2,1},$$

Within Phase 2.

$$c_{1,2} = 0 = M_{1,2}, \qquad c_{2,2} = \rho_2 = \frac{M_{2,2}}{V_2},$$

Within Fluid Phase.

$$c_f = c_{1,f} = \frac{M_{1,f}}{V_f}, \qquad c_{2,f} = \frac{M_{2,f}}{V_f}.$$

Phase Porosity

$$\phi_f = \frac{V_f}{V}, \qquad \phi_1 = \frac{V_1}{V}, \qquad \phi_2 = \frac{V_2}{V},$$

Considering only solid and fluid phases.

$$\phi_s = \phi_1 + \phi_2 = 1 - \phi_f.$$

We now can define classic variables for mixture theory, considering $p_s = p_1 = p_2$, $\mathbf{v}_s = \mathbf{v}_1 = \mathbf{v}_2$ in the solid phases and p_f, $\mathbf{v}_f$ in the fluid.

Variables.

$\mathbf{v}_f, \mathbf{v}_s$	Phase velocity	$\mathbf{u}$	Darcy velocity
p_f, p_s	Phase pressure	$\bar{p}$	Mixture pressure
ρ_f, ρ_s	Phase density	$\bar{\rho}$	Mixture density
μ_f, μ_s	Phase viscosity	$\mathbf{g}$	Gravity vector

$\rho_r = \rho_f - \rho_s$ Relative density

$\mathbf{v}_r = \mathbf{v}_f - \mathbf{v}_s$ Relative velocity

Mechanics of Flow

Darcy's Law.

$$\mathbf{u} = \phi_f \mathbf{v}_r = \phi_f(\mathbf{v}_f - \mathbf{v}_s) = -\frac{k_0 \phi_f^{2+2\Theta}}{\mu_f}(\nabla p_f - \rho_f \mathbf{g})$$

The relative permeability $k(\phi) = \phi_f^{2+2\Theta} k_0$ for simplicity

Conservation of mass

$$\frac{\partial}{\partial t}\bar{\rho} + \nabla \cdot \overline{\rho \mathbf{v}} = 0$$

Boussinesq approximation For constant and equal densities for non-buoyancy terms

$$\nabla \cdot \bar{\mathbf{v}} = \nabla \cdot (\phi_f \mathbf{v}_r + \mathbf{v}_s) = 0$$

Viscous Stokes equation.

Conservation of momentum for the deviatoric stress of the mixture is

$$\nabla \cdot \hat{\boldsymbol{\sigma}} = -\bar{\rho}\mathbf{g}$$

where

$$\hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = 2\mu_s\phi_s\Big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s\mathbf{I}\Big)$$

The symmetric gradient is defined as:

$$\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$$

The equation for this mixture stress is therefore

$$\nabla\bar{p} - \nabla \cdot \Big(2\phi_s\mu_s\Big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s\,\mathbf{I}\Big)\Big) = \bar{\rho}\mathbf{g}$$

Compaction

$$\phi_f(p_s - p_f) = -\mu_s\nabla \cdot \mathbf{v}_s,$$

Pressure Potentials. Eliminate p_s from system.

$$q_f = p_f - \rho_f g z \quad \text{and} \quad q_s = p_s - \rho_f g z,$$

$$p_f - p_s = q_f - q_s = \frac{1}{1-\phi_f}(q_f - q).$$

where $q = \bar{q}$. We can now rewrite the system with pressure potentials

$$\begin{aligned}
\phi_f \mathbf{v}_r + \frac{k_0 \phi_f^{2+2\Theta}}{\mu_f} \nabla q_f &= 0, && \text{(Darcy law)} \\
\mu_s \nabla \cdot (\phi_f \mathbf{v}_r) + \frac{\phi_f}{1-\phi_f}(q_f - q) &= 0, && \text{(conservation /compaction)} \\
\nabla q - \nabla \cdot \widehat{\boldsymbol{\sigma}}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r \mathbf{g}, && \text{(Stokes)} \\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f}{1-\phi_f}(q_f - q) &= 0 && \text{(compaction).}
\end{aligned}$$

where

$$\begin{aligned}
\widehat{\boldsymbol{\sigma}}(\mathbf{v}_s) &= 2\mu_s \phi_s \Big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I} \Big) && \text{(deviatoric stress of mixture)} \\
\mathcal{D}\mathbf{v}_s &= \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T) && \text{(symmetric gradient)}
\end{aligned}$$

Problem. When no melt, $\phi_f = 0$,

1. Darcy system degenerates when ϕ_f vanishes, loss bound on q_f.
2. Lose the conservation equation.
3. The condition number grows as $\phi_f \to 0$

Scaled variables

Define by Arbogast and Taicher [1]:

$$\tilde{\mathbf{v}}_r = \phi_f^{-\Theta}\mathbf{v}_r \quad \text{and} \quad \tilde{q}_f = \phi_f^{1/2} q_f,$$

The space H_ϕ(div)

$H_\phi(\text{div};\Omega)$ is a Hilbert space with the inner-product

$$(\mathbf{u},\mathbf{v})_{H_\phi(\text{div})} = (\mathbf{u},\mathbf{v}) + \Big(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\Big).$$

Moreover, $H(\text{div};\Omega) = H_1(\text{div};\Omega) \subset H_\phi(\text{div};\Omega)$.

New formation with scaled variables

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla(\phi_f^{-1/2} \tilde{q}_f) &= 0, \\
\mu_s \phi^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2} q) &= 0, \\
\nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r \mathbf{g}, \\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2} q) &= 0.
\end{aligned}
$$

Boundary conditions

$$
\phi_f \mathbf{v}_r \cdot \nu = \phi_f^{1+\Theta} \tilde{\mathbf{v}}_r \cdot \nu = g_r \quad \text{and} \quad \mathbf{v}_s = \mathbf{g}_s \quad \text{on } \partial\Omega,
$$

Compatibility

$$
\int_{\partial\Omega} (g_r + \mathbf{g}_s \cdot \nu)\, ds = 0.
$$

Transport of Mass and Energy

Mass Transport Equation

$$\bar{c}_t + \nabla \cdot (\overline{c\mathbf{v}} - \phi_f D \nabla c_f) = 0,$$

where bulk transport variable is defined

$$\bar{c} = \phi_f c_f + \phi_1 c_s = \frac{V_f}{V}\frac{M_{1,f}}{V_f} + \frac{V_1}{V}\frac{M_{1,1}}{V_1} = \frac{M_{1,f} + M_{1,1}}{V} = \frac{M_1}{V}$$

Standard solving approach proposed by Katz.

$$\begin{aligned} (\phi_f c_f)_t + \nabla \cdot (\phi_f c_f \mathbf{v}_f - \phi_f D \nabla c_f) &= \Gamma, \\ (\phi_1 c_s)_t + \nabla \cdot (\phi_1 c_s \mathbf{v}_s) &= -\Gamma. \end{aligned}$$

Note.

- Set $\Gamma = 0$.
- Perform flash calculation later to account for the melting term.

A different approach to solve the mass transport equation without splitting is proposed.

Definition

$$\phi_f c_f = \kappa(E, \bar{c})\,\bar{c}, \qquad \kappa := \frac{\phi_f c_f}{\bar{c}}, \qquad \kappa \in [0, 1]$$

A simplified version of transport equation with partition function is therefore proposed.

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e \bar{c} - \kappa D \nabla \bar{c}) = 0,$$

Effective Velocity

$$\mathbf{v}_e = \mathbf{v}_e^* - D(\nabla\kappa - \kappa\nabla \ln \phi_f) = \kappa\mathbf{v}_r + \mathbf{v}_s - D(\nabla\kappa - \kappa\nabla \ln \phi_f),$$

Energy Transport

$$E_t + \nabla \cdot (\overline{\rho e \mathbf{v}} - \bar{K}\nabla T) = 0,$$

where transport variable is defined

$$E = \overline{\rho e} = \phi_f \rho_f e_f + \phi_1 \rho_1 e_1 + \phi_2 \rho_2 e_2.$$

Define partition function

$$\phi_f \rho_f e_f = \lambda(E, \bar{c})\, E, \qquad \lambda := \frac{\phi_f \rho_f e_f}{E}, \qquad \lambda \in [0, 1]$$

The resulting simplified energy transport function is therefore

$$E_t + \nabla \cdot (\mathbf{v}_e^E E - \bar{K}\nabla T) = 0,$$

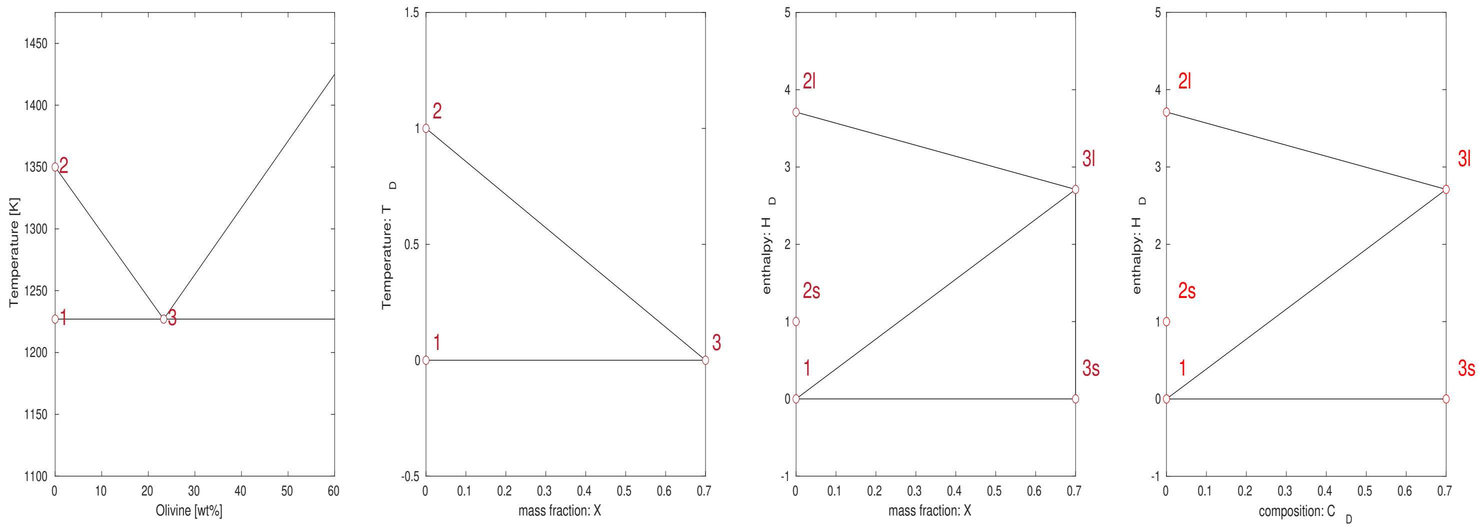
Effective velocity

$$\mathbf{v}_e^E = \lambda \mathbf{v}_f + (1 - \lambda)\mathbf{v}_s = \lambda \mathbf{v}_r + \mathbf{v}_s.$$

Phase Behavior

The flash calculation takes $\bar{c}$ and E and gives ϕ_f and T, as well as c_f and ϕ_1. A phase package is thus required.

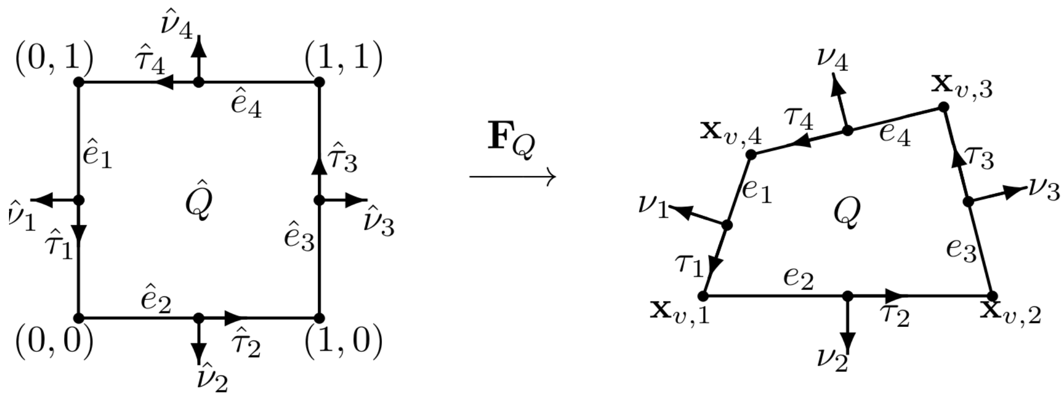
Binary Eutectic



Olivine and pyroxene are two usual immiscible cryctals in mantle melting process.

2. Computational Method

1. Conforming quadrilaterals. Should be strictly convex and not degenerate into a triangle, line or point. Maintaining logical order when adjusting grid coordinates.
2. Mixed finite element method. Second order accuracy for smooth mechanical flow.
3. Weighted essentially non-oscillatory method (WENO). Shock capturing scheme for convection diffusion equation. Third order accuracy for velocities.
4. Flash calculation. An explicit, direct parametrization of the phase diagram.



A Mixed Finite Element Method for flow

$$H_{\phi,0}(\text{div};\Omega) = \Big\{\mathbf{v} \in (L^2(\Omega))^d : \\ \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) \in L^2(\Omega),\ \phi^{1/2+\Theta}\mathbf{v}\cdot\nu = 0 \text{ on } \partial\Omega\Big\}$$

Scaled weak formulation.

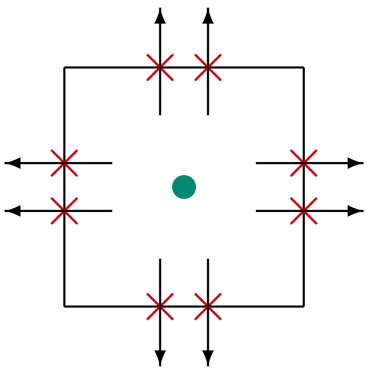
Find $\tilde{\mathbf{v}}_r \in H_{\phi,0}(\text{div})$, $\tilde{q}_f \in L^2$, $\mathbf{v}_s \in (H^1_0)^d$, and $q \in L^2/\mathbb{R}$ such that

$$\begin{aligned}
&\Big(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_r, \boldsymbol{\psi}_r\Big) - \Big(\tilde{q}_f, \phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\boldsymbol{\psi}_r)\Big) &&= 0\ \forall\,\boldsymbol{\psi}_r \in H_{\phi_f,0}(\text{div})\\
&\Big(\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r), w_f\Big) + \Big(\frac{1}{\mu_s(1-\phi_f)}(\tilde{q}_f - \phi_f^{1/2}q), w_f\Big) &&= 0\ \forall w_f \in L^2\\
&-(q, \nabla\cdot\boldsymbol{\psi}_s) + (\widehat{\boldsymbol{\sigma}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) &&= (\mathbf{G}, \boldsymbol{\psi}_s)\ \forall\,\boldsymbol{\psi}_s \in (H^1_0)^d\\
&(\nabla\cdot\mathbf{v}_s, w) - \Big(\frac{\phi_f^{1/2}}{\mu_s(1-\phi_f)}(\tilde{q}_f - \phi_f^{1/2}q), w\Big) &&= 0\ \forall w \in L^2/\mathbb{R}
\end{aligned}$$

For a polygonal domain $\Omega \subset \mathbb{R}^d \quad d = \{2, 3\}$. Assume $d = 2$ for now.

For Darcy. $\mathbb{V}_{\mathsf{D}} \times \mathbb{W}_{\mathsf{D}} \subset H_0(\mathrm{div};\Omega) \times L^2(\Omega)$.

Lowest order reduced $H(\mathrm{div})$-approximation AC space.



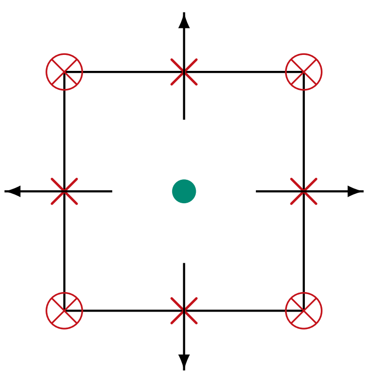
$$\mathbb{V}_{D,h}(Q) = \mathbb{V}_1^0(Q) = \mathbb{P}_1^2 \oplus \mathrm{curl}\,\mathbb{S}_1(Q)$$
$$\mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0,$$

Accuracy $\mathcal{O}(h^2)$ for $\tilde{\mathbf{v}}_r$ and $\mathcal{O}(h)$ for $\tilde{q}_f$.

DOF $8 + 1 = 9$.

For Stokes. $\mathbb{V}_{\mathsf{S}} \times \mathbb{W}_{\mathsf{S}} \subset (H_0^1(\Omega))^2 \times L^2(\Omega)$

Modified Bernardi-Raugel element.



$$\mathbb{V}_{S,h}(Q) = \mathbb{DS}_1^2(Q) \oplus \mathrm{span}\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4\}$$
$$\mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0,$$

Accuracy $\mathcal{O}(h^2)$ for $\mathbf{v}_s$ and $\mathcal{O}(h)$ for q_h

DOF $12 + 1 = 13$

Scaled mixed finite element method Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h} + \boldsymbol{g}_r$, $\tilde{q}_{f,h} \in \mathbb{W}_{D,h}$, $\mathbf{v}_{s,h} \in \mathbb{V}_{S,h} + \boldsymbol{g}_s$, and $q_h \in \mathbb{W}_{S,h}$ such that

$$
\begin{aligned}
\left(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r\right) - \left(\tilde{q}_{f,h}, \phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\boldsymbol{\psi}_r)\right) &= 0,\\
\left(\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_{r,h}), \omega_f\right) + \left(\frac{1}{\mu_s(1-\phi_f)}(\tilde{q}_{f,h} - \phi_f^{1/2} q_h), \omega_f\right) &= 0,\\
(q_h, \nabla\cdot\boldsymbol{\psi}_s) - \left(\boldsymbol{\sigma}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s\right) - ((1-\phi_f)\rho_r\mathbf{g}, \psi_s) &= 0,\\
(\nabla\cdot\mathbf{v}_{s,h}, \omega) - \left(\frac{\phi_f^{1/2}}{\mu_s(1-\phi_f)}(\tilde{q}_{f,h} - \phi_f^{1/2} q_h), \omega\right) &= 0,
\end{aligned}
$$

Lemma. There exists a unique solution to the discrete equations.

Problems.

- $\phi_f^{-1/2}$ causes problem when $\phi_f \to 0$.
- It is not locally conservative.

Locally averaged ϕ On each mesh element Q, the average of ϕ_f is $\phi_{f,Q}$, then define $\widehat{\phi}_f\Big|_Q = \phi_{f,Q}$ a constant. Using divergence theorem,

$$\int_Q \widehat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \boldsymbol{\psi}_{r,h})\, \omega_f \, dx = \int_{\partial Q} \phi_{f,Q}^{-1/2} \phi_f^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \nu \, \omega_f \, ds \\ - \int_Q \phi_{f,Q}^{-1/2} \phi_f^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \nabla \omega_f \, dx.$$

The second integral will vanish when ω_f is constant on Q. We avoid problem of $\phi_f^{-1/2}$ when $\phi_f \to 0$.

Mixed finite element problem set up Find $\tilde{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h} + \hat{\boldsymbol{g}}_r$, $\tilde{q}_{f,h} \in \mathbb{W}_{D,h}$, $\mathbf{v}_{s,h} \in \mathbb{V}_{S,h} + \hat{\boldsymbol{g}}_s$, and $q_h \in \mathbb{W}_{S,h}$ such that

$$
\begin{aligned}
\left(\frac{\mu_f}{k_0}\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r\right) - \left(\tilde{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\boldsymbol{\psi}_r)\right) &= 0,\\
\left(\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_{r,h}), \omega_f\right) + \left(\frac{1}{\mu_s(1-\phi_f)}(\tilde{q}_{f,h} - \phi_f^{1/2}q_h), \omega_f\right) &= 0,\\
(q_h, \nabla\cdot\boldsymbol{\psi}_s) - \left(\hat{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s\right) - ((1-\phi_f)\rho_r\mathbf{g}, \psi_s) &= 0,\\
(\nabla\cdot\mathbf{v}_{s,h}, \omega) - \left(\frac{\hat{\phi}_f^{1/2}}{\mu_s(1-\phi_f)}(\tilde{q}_{f,h} - \phi_f^{1/2}q_h), \omega\right) &= 0,
\end{aligned}
$$

for all test functions $\boldsymbol{\psi}_r \in \mathbb{V}_{D,h}$, $\omega_f \in \mathbb{W}_{D,h}$, $\boldsymbol{\psi}_s \in \mathbb{V}_{S,h}$, and $\omega \in \mathbb{W}_{S,h}$.

Retrieve local mass conservation With $\omega_f = \phi_{f,Q}^{-1/2}\omega_Q \in \mathbb{W}_{D,h}$ and $\omega = \omega_Q \in \mathbb{W}_{S,h}$. We define ω_Q being 1, the sum of conservation/compaction and conservation equations results in

$$
\int_Q \nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r)\,dx + \int_Q \nabla\cdot\mathbf{v}_s\,dx = 0,
$$

The unscaled Darcy potential is calculated as $q_f = \phi_f^{-1/2}\tilde{q}$. Assume that $q_{f,h} \in \mathbb{W}_{D,h}$, on Q we define

$$q_{f,h}\Big|_Q = \begin{cases} \phi_{f,Q}^{-1/2}\tilde{q}_{f,h}\Big|_Q & \text{if } \phi_{f,Q} \neq 0, \\ 0 & \text{if } \phi_{f,Q} = 0, \end{cases}$$

Relative Darcy velocity Given by $\mathbf{v}_r = \phi_f^{\ominus}\tilde{\mathbf{v}}_r$, and define the normal trace of $\mathbf{v}_{r,h} \in \mathbb{V}_{D,h}$ on each edge $e \subset \partial Q$,

$$\int_e \phi_f^{1+\ominus}\tilde{\mathbf{v}}_{r,h}\cdot\nu\, ds = \int_e \phi_f \mathbf{v}_{r,h}\cdot\nu\, ds.$$

Using a weighted L^2 projection on the edge e, denoted as $P_1^{\phi_f} : L^2(e) \to \mathbb{P}_1(e)$ and defined by the requirement that for $f \in L^2(e)$, $P_1^{\phi_f} f \in \mathbb{P}_1(e)$ and

$$\int_e \phi_f (f - P_1^{\phi_f} f)\,\ell\, ds = 0 \quad \forall \ell \in \mathbb{P}_1(e).$$

Then

$$\mathbf{v}_{r,h}\cdot\nu = P_1^{\phi_f}(\phi_f^{\ominus}\tilde{\mathbf{v}}_r\cdot\nu),$$

Auxiliary functions In construction of the direct serendipity space, we start by defining auxiliary functions.

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{vi}) \cdot \nu_i, \quad i = 1, 2, 3, 4,$$

Then define

$$R_{13}(\mathbf{x}) = \frac{\lambda_1 - \lambda_3}{\lambda_1 + \lambda_3}, \quad R_{24}(\mathbf{x}) = \frac{\lambda_2 - \lambda_4}{\lambda_2 + \lambda_4},$$

and

$$R_1(\mathbf{x}) = \tfrac{1}{2}(1 - R_{13}), \quad R_3(\mathbf{x}) = \tfrac{1}{2}(1 + R_{13}),$$
$$R_2(\mathbf{x}) = \tfrac{1}{2}(1 - R_{24}), \quad R_4(\mathbf{x}) = \tfrac{1}{2}(1 + R_{24}).$$

Function R_i is 1 on corresponding edge e_i, 0 on the opposite edge, and arbitrary on two neighbours.

$$\mathbb{S}_1(Q) = \text{span}\Big\{\lambda_2\lambda_4 R_{13},\ \lambda_1\lambda_3 R_{24}\Big\},$$

Resulted $\mathbb{DS}_1(Q)$ space

$$\mathbb{DS}_1(Q) = \mathbb{P}_1 \oplus \mathbb{S}_1(Q),$$

Defined on the given quadrilateral Q.

The modified Bernardi-Raugel finite element is therefore defined as following:

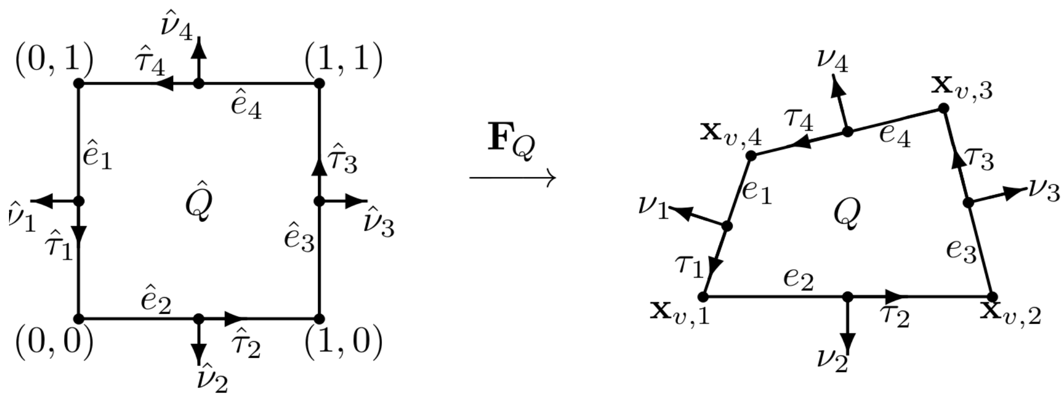
$$\mathbb{V}_{S,h}(Q) = \mathbb{DS}_1^2(Q) \oplus \text{span}\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4\}$$
$$\mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0,$$

Supplemental functions

Four supplemental functions are defined modulo 4 by

$$\mathbf{b}_i(\mathbf{x}) = \frac{\varphi_i(\mathbf{x})}{\varphi_i(\mathbf{x}_i)}\nu_i, \quad \mathbf{x}_i = \tfrac{1}{2}(\mathbf{x}_{v,i-1} + \mathbf{x}_{v,i}), \quad \varphi_i = \lambda_{i+1}\lambda_{i+3}R_i,$$

We remark that the scalar function φ_i lie in second order serendipity space $\mathbb{DS}_2(Q)$.



Notations.

- ν_i Unit outer normal.
- τ_i Unit tangent vector.
- e_i Edge middle point.
- $\mathbf{x}_i$ Vertex points.

Advantages

- Fit complex geometry.
- Require less element than triangular mesh.

Note.

1. When $d = 2$, the convergence of the linear serendipity finite element space does not degenerate on quadrilateral mesh.
2. Direct serendipity space defined directly on quadrilateral mesh without mapping from reference rectangular.

WENO-AO Method for Transport

Transport system equation

$$U_t + \text{div}[F(U) - A(U)\text{grad}U] = 0,$$

where the transport variable is defined $U = (\bar{c}, E)$ row-wise.

Flux and diffusion coefficient matrices

$$F(U) = F\begin{pmatrix} \bar{c} \\ E \end{pmatrix} = \begin{pmatrix} \mathbf{v}_e^T \bar{c} \\ (\mathbf{v}_e^E)^T E \end{pmatrix} \quad \text{and} \quad A(U) = \begin{bmatrix} \kappa D & 0 \\ \frac{\partial T}{\partial \bar{c}} & \frac{\partial T}{\partial E} \end{bmatrix},$$

In terms of the average of U over the element Q,

$$U_Q(t) = \begin{pmatrix} \bar{c}_Q(t) \\ E_Q(t) \end{pmatrix}, \quad \bar{c}_Q(t) = \frac{1}{|Q|} \int_Q \bar{c}(x,t)\,dx,$$

and

$$E_Q(t) = \frac{1}{|Q|} \int_Q E(x,t)\,dx,$$

Basic finite volume scheme

$$\begin{aligned} U_{Q,t}(t) &= -\frac{1}{|Q|}\int_Q \mathrm{div}\Big[F(U) \ - A(U)\mathrm{grad}U\Big]\,dx \\ &= -\frac{1}{|Q|}\int_{\partial Q}\Big[F(U)\nu - A(U)\nabla U\cdot\nu\Big]\,ds, \end{aligned}$$

Remarks.

- Value of U on the edge will be constructed with WENO-AO method.
- Boundary integral will be approximated with appropriate Gauss quadrature.
- Lax-Friedrichs type numerical flux will be used as stabilizer.
- Implicit time stepping (Radau-IIA method) will be used.

A WENO-AO Reconstruction

The Weighted essentially non-oscillatory (WENO) method is designed for the degenerate advection-diffusion equation. The WENO-AO method stands for WENO method with adaptive reconstruction order. This adaptive order version allows accuracy drops to the highest possible of accuracy near the discontinuity rather than the base level.

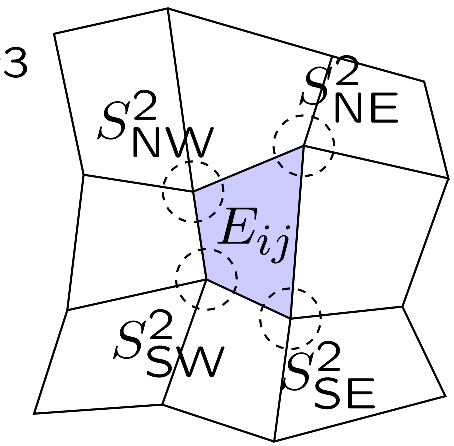
Twp types of flux reconstruction

- WENO-AO(3,2) reconstruction for convective flux,
- WENO-AO(4,3) reconstruction for diffusive flux.

Remarks.

- Assume 2d reconstruction for this presentation.
- Diffusive flux needs to be fourth order accuracy for a third order reconstruction of the derivative.
- Base second order accuracy in consistency with mixed finite element method.

WENO-AO(3,2) Reconstruction



The WENO-AO(3,2) reconstruction of point value S^3
u is given,

$$u(x) \approx R_i^{3,2}(x) = \frac{\tilde{\alpha}}{\alpha}[P^3(x) - \sum_{k \in I} \beta_k P_k^2(x)] + \sum_{k \in I} \tilde{\beta}_k P_k^2$$

Nonlinear weights

$$\widehat{\alpha} = \frac{\alpha}{(\epsilon + \sigma_{P^3})^\tau}, \quad \widehat{\beta}_k = \frac{\beta_k}{(\epsilon + \sigma_{P_k^2})^\tau}$$

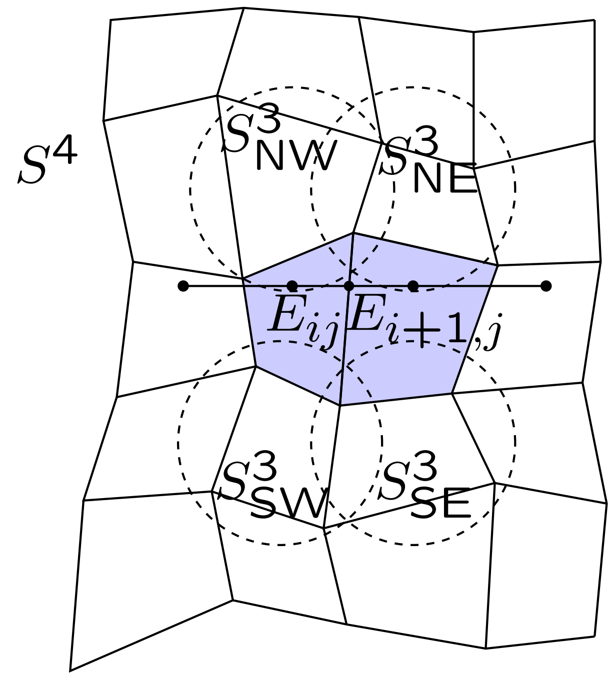
$$\tilde{\alpha} = \frac{\alpha}{\widehat{\alpha} + \sum_{l \in I} \widehat{\beta}_l}, \quad \tilde{\beta}_k = \frac{\beta_k}{\widehat{\alpha} + \sum_{l \in I} \widehat{\beta}_l}$$

Smoothness Indicator

$$\sigma_{p^r} = \sum_{l=1}^{r-1} \int_{E_i} \Delta x_i^{2l-1} (\frac{d^l}{dx^l} P^r(x))^2 dx$$

Because WENO-AO(4,3) is used in reconstruction of diffusive term. The fourth order reconstruction should be reconstructed in a symmetric manner to avoid directional bias.

$$b_x(x) \approx R_i^{4,3}(x) = \frac{\tilde{\alpha}}{\alpha}[Q^{4'}(x) - \sum_{k \in I} \beta_k Q_k^{3'}(x)] + \sum_{k \in I} \tilde{\beta}_k Q_k^{3'}$$



The details of WENO-AO(3,2) and WENO-AO(4,3) reconstruction can be found in reference [2].

A Lax-Friedrichs Type Numerical Flux

Normal flux and Jacobian

$$F(U)\nu = \begin{pmatrix} \mathbf{v}_e \cdot \nu\, \bar{c} \\ \mathbf{v}_e^E \cdot \nu\, E \end{pmatrix}$$

and

$$\frac{\partial F(U)\nu}{\partial U} = \begin{bmatrix} \frac{\partial(\mathbf{v}_e \cdot \nu\, \bar{c})}{\partial \bar{c}} & \frac{\partial(\mathbf{v}_e \cdot \nu\, \bar{c})}{\partial E} \\ \frac{\partial(\mathbf{v}_e^E \cdot \nu\, E)}{\partial \bar{c}} & \frac{\partial(\mathbf{v}_e^E \cdot \nu\, E)}{\partial E} \end{bmatrix} \approx \begin{bmatrix} \mathbf{v}_e \cdot \nu & 0 \\ 0 & \mathbf{v}_e^E \cdot \nu \end{bmatrix}.$$

Lax-Friedrichs flux

Replace this normal flux with $\widehat{F}(U^-, U^+)$, where U^- and U^+ are one-sided limits of the approximate solution.

$$\widehat{F}(U^-, U^+) = \frac{1}{2}\Big[F(U^-)\nu + F(U^+)\nu - \alpha_{\mathsf{LF}}(U^+ - U^-)\Big], \tag{1}$$

where

$$\alpha_{\mathsf{LF}} = \begin{bmatrix} \|\mathbf{v}_e \cdot \nu\|_{L^\infty} & 0 \\ 0 & \|\mathbf{v}_e^E \cdot \nu\|_{L^\infty} \end{bmatrix}. \tag{2}$$

We discretize time into levels $t^0 < t^1 < t^2 < \cdots$ and set $\Delta t^{n+1} = t^{n+1} - t^n$. The time derivative will be integrated over the interval $[t^n, t^{n+1}]$ using a third order Runge-Kutta time stepping method.

Radau-IIA method A third order implicitt Runge-Kutta method. A Butcher Tableau for this mehtod is presented.

$$
\begin{array}{c|cc}
1/3 & 5/12 & -1/12 \\
1 & 3/4 & 1/4 \\
\hline
 & 3/4 & 1/4
\end{array}
$$

Remark.

Radau-IIA method is simple and L-stable at the same time.

Two Different Algorithms

Algorithm 1: Sequential solution

Let U^0 be given as initial data.

Initialize:

1. Given U^0, compute W^0 by the flash calculation.
2. Given ϕ_f^0 in W^0, solve the Darcy-Stokes system for V^0.

for Time level $n = 0, 1, \ldots, n_{\text{max}} - 1$ **do**

1. Given W^n and V^n, define
$$\kappa^n = \frac{\phi_f^n c_f^n}{\bar{c}^n} \qquad \text{and} \qquad \lambda^n = \frac{\phi_f^n \rho_f e_f^n}{E^n},$$
and then solve the transport system for U^{n+1}.
2. Given U^{n+1}, compute W^{n+1} by the flash calculation.
3. Given ϕ_f^{n+1} in W^{n+1}, solve the Darcy-Stokes system for V^{n+1}.

end

Notation.

- $V^n = (\mathbf{v}_{h,r}^n, \mathbf{v}_{h,s}^n, q_{h,f}^n, q_h^n)$ denote the solution to the Darcy-Stokes system at the same time.
- $W^n = (\phi_f^n, c_f^n, T^n, \phi_1^n)$ denote the output of the flash calculation at t^n.

Algorithm 2: Iteratively coupled solution

Let U^0 be given as initial data.

Initialize W^0 and V^0 as in Algorithm **??**.

for Time level $n = 0, 1, \ldots, n_{\text{max}} - 1$ **do**

Set $U^{n,0} = U^n$, $V^{n,0} = V^n$, and $W^{n,0} = W^n$.

for Newton iteration $\ell = 0, 1, \ldots$ until convergence (or $\ell_{\text{max}} - 1$) **do**

1. Given $W^{n,\ell}$ and $V^{n,\ell}$, define

$$\kappa^{n,\ell} = \frac{\phi_f^{n,\ell} c_f^{n,\ell}}{\overline{c}^{n,\ell}} \quad \text{and} \quad \lambda^{n,\ell} = \frac{\phi_f^{n,\ell} \rho_f e_f^{n,\ell}}{E^{n,\ell}},$$

and then solve the transport system for $U^{n,\ell+1}$. Evaluate U and W at their proper times within the equation (i.e., at time t^n, or at time t^{n+1} using the (n, ℓ) values, or by interpolating these two values to an intermediate time).

2. Given $U^{n,\ell+1}$, compute $W^{n,\ell+1}$ by the flash calculation.
3. Given $\phi_f^{n,\ell+1}$ in $W^{n,\ell+1}$, solve the Darcy-Stokes system for $V^{n,\ell+$

end

Set $U^{n+1} = U^{n,\ell+1}$, $V^{n+1} = V^{n,\ell+1}$, and $W^{n+1} = W^{n,\ell+1}$.

end

Algorithm 1.
U^{n+1} is computed using fixed in time values for the effective velocities $\mathbf{v}_e$ and $\mathbf{v}_e^E$, pre-calculated from the Darcy-Stokes part, and the partition functions κ and λ. These quantities are only updated before commencing to the next time step.

Algorithm 2.
Use the current information within a monolithic quasi-Newton solve. It still reduces to solving the three subproblems. We simply omit the Newton derivative couplings between the subproblems. However, we will also explore improving the partition function approach by linearizing them, i.e., for κ,

$$\kappa = \kappa^{n,\ell} + d\kappa^{n,\ell}(U^{n,\ell+1} - U^{n,\ell}),$$

and similarly for λ. This will require computing derivatives of κ and λ in the flash package.

3. Application to Mantle Dynamics

Simulation of partial melting beneath mid-ocean ridges.

1. The high resolution transport scheme proposed by us will provide a detailed simulation result of melt transport and segregation.
2. Lead to a better understand of the dynamics of mid-ocean ridges and hotpots by resolving the small scale dynamics of these processes in solitary waves.
3. The possible degeneracies in the porosity will be well represented and the survial of distinct geochemical signatures in the decompaction channel will also be invested.

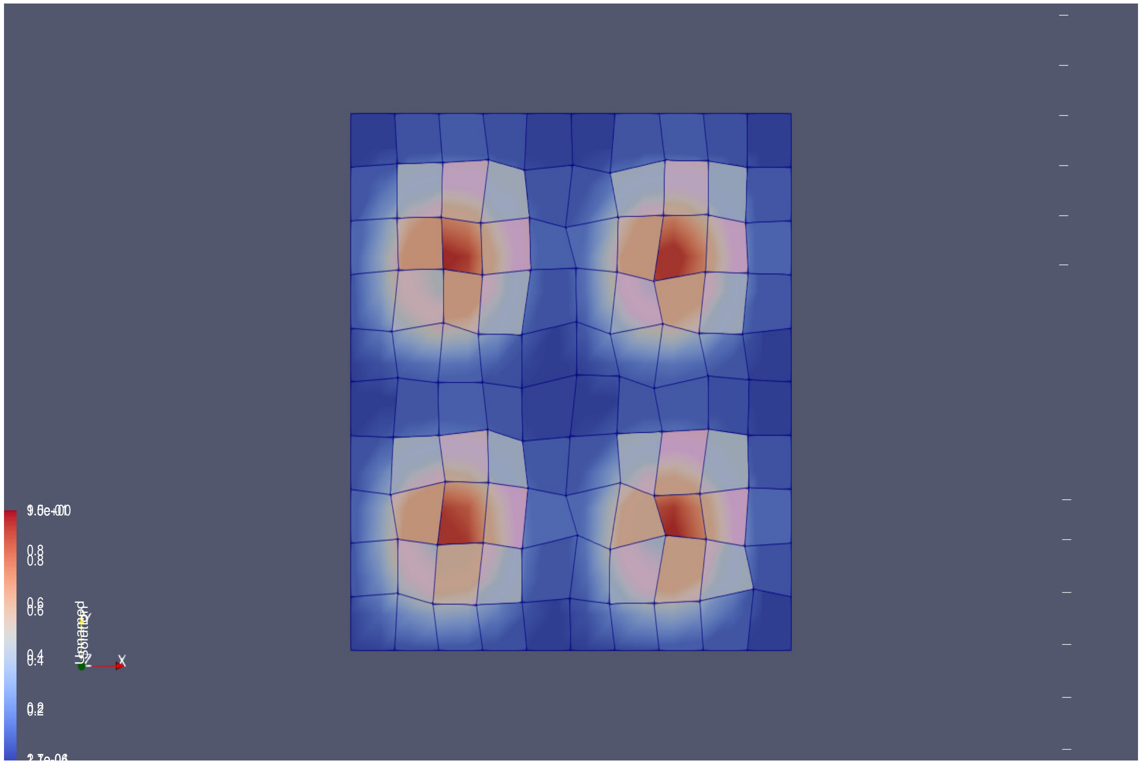
4. Some Preliminary Numerical Results

Code Details

- Parallelization will be implemented with data management provided by PETSc.
- Nonlinear solver and preconditioner will be controlled through PETSc functions.
- Simple linear solver will be supplied by LAPACK.
- Output will be HDF5 format with the help of CGNS (CFD General Notation System).

Visualization of Quadrilateral mesh with CGNS

Example output of quadrilateral mesh.



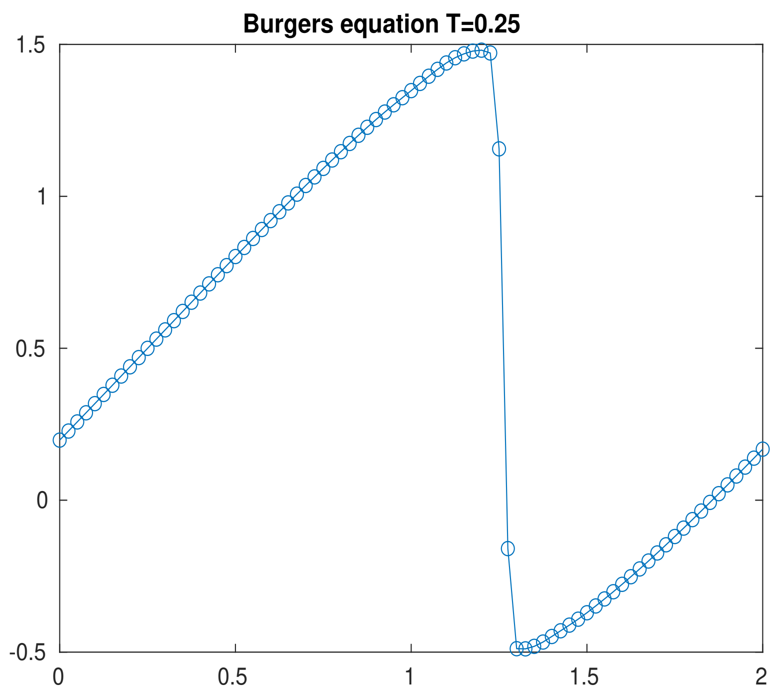
Remarks.

- Can fit complex geometry.
- Require fewer elements than triangular mesh.
- Require mesh quality check.
- Constructed by randomly distortion.

A simple 1D Burgers equation with periodic boundary condition has been solved with the PETSc-based code. The implicit WENO3 method was used here.

$$u_t + \frac{u^2}{2} = 0$$

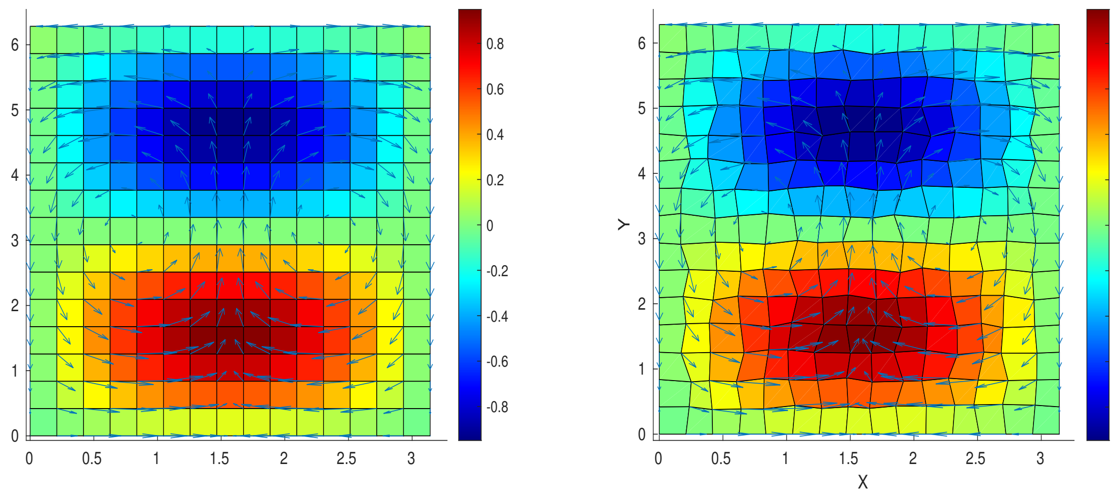
Computed numerical result



Analytical result

$$u = \cos(x)\sin(y), \qquad v = -\sin(x)\cos(y),$$
$$\frac{\partial p}{\partial x} = \cos(x)\sin(y), \quad \frac{\partial p}{\partial y} = \sin(x)\cos(y).$$

Computed numerical result



The computed numerical result is compared to this exact solution by calculating the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.

Rectangular mesh

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\text{res}_p)$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\text{res}_{du})$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Quadrilateral mesh

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E−1	1.074E−1	2.567E−2	6.266E−3
$L^2(\text{res}_p)$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\text{res}_{du})$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Remark.

- The order of accuracy does not degenerate on quadrilateral mesh as expected.
- Second order accuracy for velocity and pressure.
- First order accuracy for derivatives of the velocity.

5. Summary

- Area A:

  1. A new and simpler local mass conservative mixed finite element scheme for the Darcy-Stokes system.
  2. A modified Bernardi-Raugel element equiped with direct serendipity space. The stability and convergence of this new element will be further studied.
  3. Analyze propagation of errors within the discrete coupled system.
  4. Analyze mass conservation in the context of the complete coupled system of equations.

- Area B:

  1. A PETSc implementation of the proposed algorithm. Parallel running will be enabled in the solver.
  2. The suitability of the linearized formulation of κ and λ will be studied.
  3. The suitability of Lax-Friedrichs type numerical flux.
  4. First implement of the implicity WENO-AO method in full scale mantle simulation.

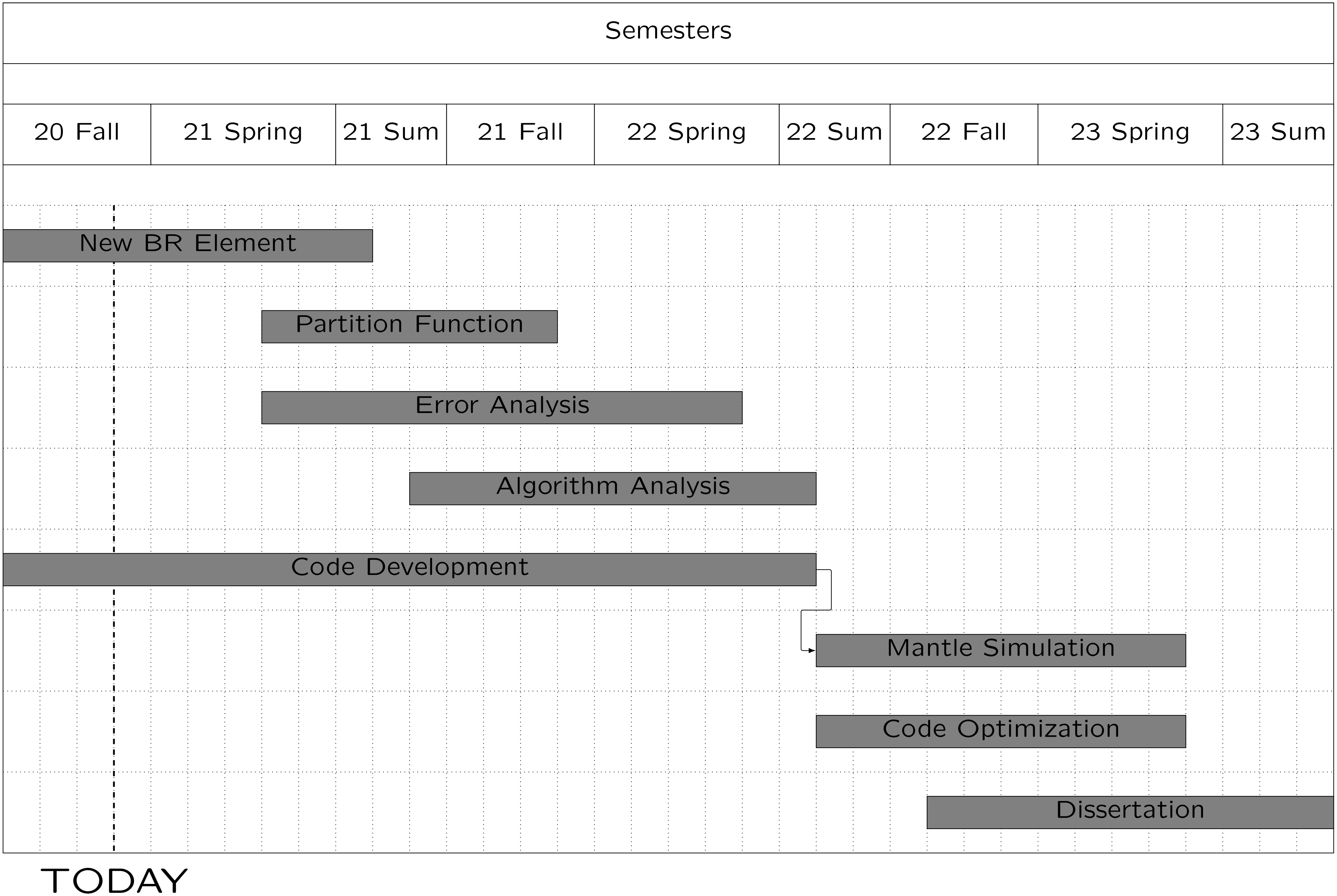
- Area C
    1. New partition function method will be compared with standard approach proposed by Katz.
    2. New local mass conservation scheme will be studied from a physical point of view.
    3. Expect to see a improved approximation of the dynamics of partial melting in simulatins of melt extraction beneath mid-ocean ridge.

Future Work and Timeline

Future Work

1. Continue studying new mixed finite element space.
2. Develop a feasible code.
3. Perform numerical simulation of real world problems.

Timeline Gantt Chart



Reference

1. T. Arbogast and A. L. Taicher, *A linear degenerate elliptic equation arising from two-phase mixtures,* SIAM J. Numer. Anal. 54:5 (2016), pp. 3105-3122. DOI 10.1137/16M1067846.
2. T. Arbogast, C.-S. Huang, and X. Zhao, *Accuracy of WENO and adaptive order WENO reconstructions for solving conservation laws*, SIAM. J. Numer. Anal., 56 (2018), pp. 1818–1847. DOI 10.1137/17M1154758.
3. T. Arbogast, M. A. Hesse, and A. L. Taicher, *Mixed methods for two-phase Darcy-Stokes mixtures of partially melted materials with regions of zero porosity,* SIAM J. Sci. Comput. 39:2 (2017), pp. B375–B402. DOI 10.1137/16M1091095.